VOLLEY-G5-ACR1

Mass properties, material classes and fabrication boundary

OUTLINE / PRELIMINARY ENGINEERING — dimensions in mm; no general manufacturing tolerance assigned

HOME native assembly: 387 parts; loaded 197.556720 kg = dry 149.556720 + payload 48.000000.

Represented dry CG (mm): X 628.361876; Y -4.213144; Z 267.524868.

Loaded native CG (mm): X 581.198863; Y -3.189484; Z 337.372183.

Retained sled with pallet: 13.127806 kg. Mission device: 156.556720 kg; loaded device: 204.556720 kg.

Extra 7.000 kg power/control allowance has NO assigned position and is excluded from CAD CG/inertia.

Represented dry inertia about its own CG [kg·m²], full tensor, source sign convention:

[29.675715228 0.060095224 12.702423066]
[0.060095224 62.806994278 0.035579787]
[12.702423066 0.035579787 54.885372840]

Materials are assigned density classes; no alloy temper, coating, bond specification or supplier certification is assigned.

Aluminium 2700; steel 7850; copper 8960; NdFeB 7500; polymer 1410; insulator 1800 kg/m³.

Catalog/payload envelope mass distributions remain approximate; see exact per-group source boundaries.

BOM_HOME.csv contains all native part labels, assigned masses, CG coordinates, volumes and source notes.

MASS_PROPERTIES.json retains all eight native-state accepted group CG/inertia records.

PRELIMINARY FABRICATION ROUTE

Frame/pallet: stock machining and joint/inspection planning remain to be detailed from the owned solid model.

Copper phases: preserve connected topology, minimum return section and insulation gaps; joining/insulation process unselected.

Guides, screws, bearings, brakes and actuators: source envelope/reference only where catalog interfaces are unselected.

No fastener torque, interference fit, vacuum material compatibility or supplier acceptance is invented by these sheets.